BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Multiple Choice Questions)

> Note: Q.1 Choose the correct answer for each from the given options:

Q1. Albumins are the type of _____.

(a) Conjugated Proteins
(b) Simple Protein
(c) Derived Proteins
(d) Monasaccharides

Q2. Which of the following polymer is used in the manufacture of chewing gurms?

(a) Bakelite
(b) PVC
(c) PVA
(d) Poly ethylene

Q3. Which of the following is an isotope of hydrogen?

(a) Protium
(b) Watergas
(c) Atomic Hydrogen
(d) Ionic hybride

Q4. Which of the following element have variable oxidation states?

(a) Na
(b) F
(c) Be
(d) Mn

Q5. Hydrogen shows partial resemblance with the elements of _____ groups.

(a) IA, IIA and IIIA
(b) IA, IVA and VIIA
(c) IIIA, IVA and VIIA
(d) IA, IIA and VIIIA

Q6. Formula of plaster of paris is _____.

(a) $CaSO_4, 2H_2O$
(b) $CaSO_4, 3H_2O$
(c) $CaSO_4, ½ H_2O$
(d) $CaSO_4$

Q7. Which method is used for the purification of Bauxite containing excess of Fe2O3?

(a) Serpek's Method
(b) Bacyers Method
(c) Hall's Method
(d) Contact Method

Q8. Which of the following compounds is homocyclic?

(a) Furan
(b) Pyridine
(c) Benzene
(d) Thiophene

Q9. Elements of the same period have same number of _____.

(a) Protons
(b) Electrons
(c) Neutrons
(d) Shells

Q10. Isomers have different _____.

(a) Molecular Formulae
(b) Crytalline Structure
(c) Structural Formulae
(d) Refractive Index

Q11. The first four members of alkane series are _____.

(a) All liquids
(b) All gases
(c) Two liquids and two gases
(d) Three liquids and one gas

Q12. The general formula of Grignord's reagents is _____.

(a) $R - X$
(b) $R2CH - X$
(c) $R - Mg - X$
(d) R3CX

Q13. Which of the following give addition reactions?

(a) CH3OH
(b) HCHO
(c) CH4
(d) CH3Cl

Q14. Which type of compounds will be formed by treating Grignord's reagent with Ketones?

(a) Primary Alcohols
(b) Secondary Alcohols
(c) Tertiary Alcohols
(d) Alkenes

Q15. An organic compound containing carbonyl group and hydroxyl agroup is known as _____.

(a) Alcohol
(b) Aldehyde
(c) Ketone
(d) Carboxylic Acid

Q16. Which one of the following gases shows acidic properties?

(a) Acetylene
(b) Methane
(c) Ethane
(d) Butane

Q17. The catalyst used in hydrogenation reaction is _____.

(a) KMnO4
(b) K2Cr2O7
(c) Ni/pt/pd
(d) V2O5

Q18. Which of the following is a polysaccharide?

(a) Glucose
(b) Fructose
(c) Maltose
(d) Cellulose

Q19. Mendleev was a _____ chemist.

(a) British
(b) German
(c) Russian
(d) French

Q20. Which of the following contains ionic bonds?

(a) CH4
(b) NaH
(c) H2O
(d) NH3

Q21. NaOH + NH4Cl →

(a) NH4OH + NaCl
(b) NH3 + H2O + NaCl
(c) NaH + H2O
(d) Both (a) and (b)

Q22. Why Graphite conducts electricity and Dimond does not?

(a) In Graphite each carbon possess one single electron
(b) In Graphite each carbon possess all paired electrons
(c) Due to black colour of graphite
(d) Because graphite is soft one

Q23. In which of the following the complex ion is an anion?

(a) [Cr (NH3)4 Cl2]Cl
(b) [Co (NH3)6] Cl3
(c) K3 [Co (NO2)6]
(d) [pt (NH3)2 Cl4]

Q24. Bituminous coal on _____ produces coke.

(a) Destructive Distillation
(b) Fractional Distillation
(c) Cracking
(d) Polymerization

Q25. Which of the following reactions of benzene is not an electrophilic substitution?

(a) Nitration
(b) Sulphonation
(c) Hydrogenation
(d) Friedal − craft's reactions

Q26. Hybridization of carbon in acetylene is _____

(a) SP^3
(b) SP
(c) SP^2
(d) Both (a) and (b)

Q27. The solvent which favours SN1 reaction is _____.

(a) Polar
(b) Organic
(c) Non Polar
(d) Both (b) and (c)

Q28. Ketones result by the oxidation of _____.

(a) Primary Alcohols
(b) Secondary Alcohols
(c) Tertiary Alcohols
(d) Both (a) and (c)

Q29. The product obtained by the fermentation of Glucose is _____.

(a) $C_2H_5OH + CO_2$
(b) $CH_3OH + CO_2$
(c) $CO + H_2O$
(d) $NaCl + H_2O$

Q30. Which of the following vitamins are water soluble?

(a) A, B, C, D
(b) A, D, C, K
(c) D, C, K
(d) B and C

Q31. Which of the following is an electrophile?

(a) OH
(b) CH
(c) NO_2
(d) Both (a) and (b)

Q32. Which of the following is not a ligand?

(a) CH_3
(b) H_2O
(c) NH_3
(d) $NH_2-CH_2-CH_2-NH_2$

SECTION – B (SHORT ANSWER)

> Note: Answer any SEVEN of the following questions. Each question carries 05 marks.

Q2. What is formaldehyde? How can it be prepared?

Q3. What do you know about hydrides? Define any two types.

Q4. Explain why: (i) Cl- ion is larger in size than Cl atom (ii) HI is more acidic than HBr (iii) Ionic radius of Mg is smaller than that of Na (iv) HNO_3 is strong oxidizing agent (v) d – block elements form coloured compounds

Q5. Define amino acids with examples.

Q6. How HNO_3 is prepared?

Q7. Define plastics with examples.

Q8. Discuss E1 and E2 reactions with examples.

Q9. Give any two methods of preparation of CH_4.

Q10. Discuss any two chemical reactions of acetic acid.

Q11. What do you know about the structure of acetylene?

SECTION – C (DESCRIPTIVE ANSWER)

> Note: Answer any TWO of the following questions. Each question carries 09 (5 + 4) marks.

Q12. (a) Name the following compounds by I.U.P.A.C system. (i) OH − CH2 − CH2 − OH (ii) CH3 − CH − C = O (Cl, OH below) (iii) K3 [Fe (CN)6] (v) (CH3)3. C − CH2 − CH (Cl above, Cl below) (v) [benzene ring diagram with NO2, OH, NO2 substituents] (b) Draw the structure of the following. (i) Methyl Acetate (ii) Catechol (iii) Methoxy Ethane (iv) Tertiary Butyl Alcohol

Q13. (a) How alumina is purified from bauxite? Explain with examples. (b) Discuss the structure of ethane.

Q14. (a) How caustic soda is prepared by castner – kellner's process? Discuss the merits and demerits of the method. (b) Complete the following reactions: (i) Cl2 + H2S → (ii) C + H2SO4 → (iii) CH3I + Na → (Ether) (iv) CH3CHO + CH3CHO → (KMnO4)